

Superstore Sales Data

Import, Power Query & Data Cleaning

Week 1 – Documentation Report
Prepared in Power BI Desktop – Power Query Editor

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Tool	Microsoft Power BI Desktop
Dataset	Superstore Sales Dataset

This report documents the Week 1 preparation of the Superstore sales dataset. The data was imported into Power BI, inspected in Power Query, checked for data-quality issues, standardized in selected categorical columns, and enhanced with Year, Quarter, and Month attributes derived from Order Date.

1. Objective

The objective of this exercise is to prepare the raw Superstore sales dataset for later analysis and dashboard development. The Week 1 workflow focuses on importing the data, validating its structure and data types, checking for errors and missing values, standardizing selected text fields, and creating useful date attributes from Order Date.

2. Tasks Covered

1. Load the Superstore dataset into Power Query Editor.
2. Promote the first row as column headers and verify the table structure.
3. Verify and correct relevant column data types.
4. Check for missing values and errors using column profiling.
5. Verify record uniqueness and inspect for duplicate records.
6. Apply Trim to Category, Sub-Category, and Segment.
7. Extract Year, Quarter, and Month from Order Date.
8. Close & Apply the completed Power Query transformations.

3. Import and Initial Inspection

The Superstore dataset was loaded into Power BI Desktop and opened in Power Query Editor. The first row was promoted as the column headers. The table structure and column profiling were inspected before applying transformations.

The Order Date and Ship Date fields were handled as date values. Change Type with Locale was used where necessary so the source date format was interpreted correctly before date-based transformations were performed.

4. Data Type Verification

Relevant fields were assigned appropriate data types so that numerical calculations and date transformations would work correctly.

Field	Data type
Row ID	Whole Number
Order ID	Text
Order Date	Date
Ship Date	Date
Quantity	Whole Number
Sales	Decimal Number
Discount	Decimal Number
Profit	Decimal Number

5. Missing Values, Errors and Record Uniqueness

Column profiling was used to inspect the dataset for errors and empty values. The checked columns showed 100% valid values, with 0% errors and 0% empty values.

Record uniqueness was checked using Row ID as the record identifier, and no exact duplicate rows were identified during the inspection.

6. Text Standardization

The Trim transformation was applied to Category, Sub-Category, and Segment. This removes unnecessary leading or trailing spaces and helps keep categorical values consistent for grouping, filtering, and visualization.

The text-cleaning step was limited to the categorical fields required for the Week 1 task.

7. Extracting Date Attributes

The Order Date column was used to create three additional attributes required for the Week 1 task: Year, Quarter, and Month. These fields will support time-based grouping and analysis in the later dashboard.

Year: Add Column → Date → Year → Year.

Quarter: Add Column → Date → Quarter → Quarter of Year.

Month: Add Column → Date → Month → Month.

The screenshot displays the Power Query Editor interface for a dataset named 'superstore'. The main area shows a table with columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, and Customer Name. Each column has a data type and a status (Valid, Error, Empty) with corresponding percentages. The 'Order Date' column is highlighted. The 'Applied Steps' pane on the right shows a list of steps: Source, Promoted Headers, Changed Type with Locale, Changed Type with Locale1, Changed Type, and Trimmed Text. The 'Trimmed Text' step is selected, indicating that text has been cleaned in the specified columns.

Column	Data Type	Valid	Error	Empty
Row ID	Text	100%	0%	0%
Order ID	Text	100%	0%	0%
Order Date	Date	100%	0%	0%
Ship Date	Date	100%	0%	0%
Ship Mode	Text	100%	0%	0%
Customer ID	Text	100%	0%	0%
Customer Name	Text	100%	0%	0%

Figure 1. Power Query Editor showing the cleaned table and Applied Steps.

Order Date]	Year	Quarter	Month
100%	Valid	100%	Valid
0%	Error	0%	Error
0%	Empty	0%	Empty
5.5512	2023	1	1
-5.487	2023	1	1
4.2717	2023	1	1
-64.7748	2023	1	1
4.884	2023	1	1
746.4078	2023	1	1
1.4796	2023	1	1
5.2398	2023	1	1
274.491	2023	1	1
0.3112	2023	1	1
3.0084	2023	1	1
9.3312	2023	1	1
204.1092	2023	1	1
113.6742	2023	1	1
-53.7096	2023	1	1
-18.2525	2023	1	1
1.168	2023	1	1
9.75	2023	1	1
21.2954	2023	1	1
1.3583	2023	1	1
3.0814	2023	1	1
3.9294	2023	1	1
87.3504	2023	1	1
6.567	2023	1	1

Figure 2. Year, Quarter, and Month columns created from Order Date.

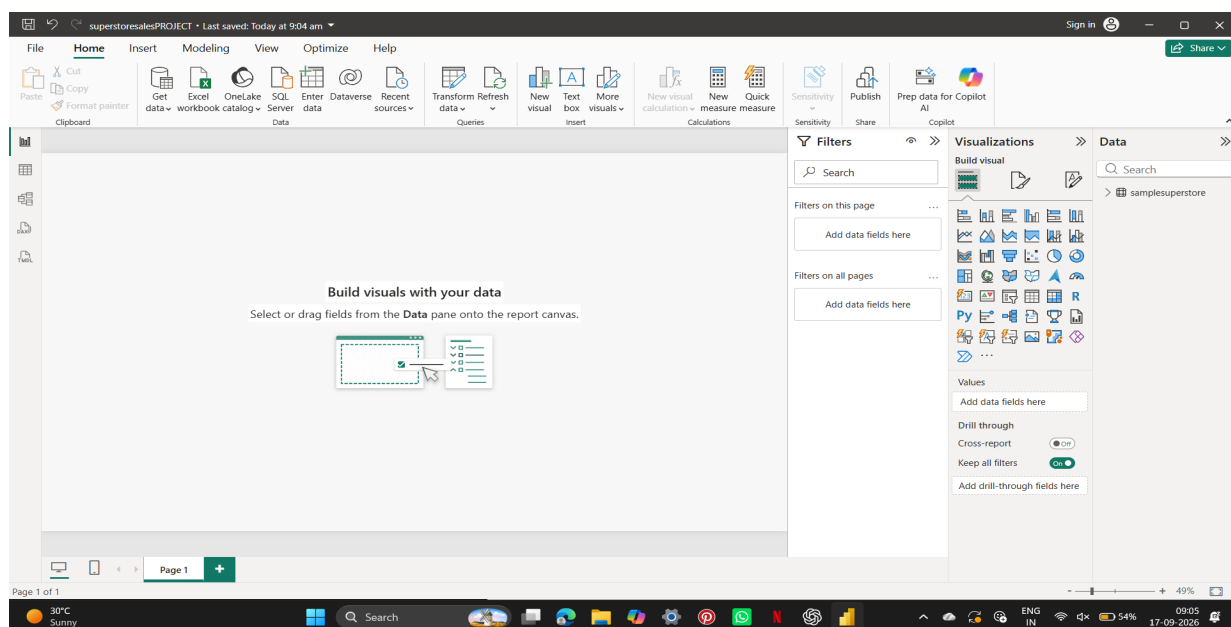


Figure 3. Power BI report page after Close & Apply.

8. Close & Apply

After completing the required transformations, Close & Apply was selected in Power Query Editor. The changes were loaded back into the Power BI report for use in subsequent modeling, DAX, and dashboard tasks.

9. Conclusion

The Week 1 Superstore data-preparation exercise was completed in Power BI Desktop using Power Query Editor. The dataset was imported and inspected, data types were handled correctly, data quality was checked, selected categorical fields were standardized, and Year, Quarter, and Month were derived from Order Date. The resulting table is prepared for the Week 2 activities involving data modeling, DAX measures, and dashboard development.

WEEK 1 — COMPLETED